Proprietary US Inflation Forecasts

Executive summary. We compare our proprietary bottom-up US inflation model and our Machine Learning (ML) model with two demanding external benchmarks: the professional Consensus and Kalshi's market-implied CPI forecast. The fair comparison is conducted separately for headline and core CPI, month on month and year on year, on identical four-model panels beginning no earlier than January 2023; every result reports its observation count. Our analysis is based on real-time data. **Ten days before release, bottom-up has the lowest pooled mean absolute error (0.070), ahead of Consensus (0.077), ML (0.082) and Kalshi (0.097). It also has the highest exact one-decimal rate and the smallest worst unrounded miss (0.250 percentage point). ML's clearest point-forecast advantage is headline year-on-year CPI, where its MAE of 0.084 beats both Consensus (0.090) and Kalshi (0.092).**

Kalshi becomes more accurate as the release approaches, but it does not erase the pooled proprietary advantage. On identical six-forecast panels, its pooled MAE declines from 0.097 at ten days to 0.089 at one day and 0.089 at one hour. Those near-release errors remain above bottom-up (0.070), Consensus (0.077) and ML (0.082). Kalshi catches up most clearly in month-on-month CPI; the proprietary edge survives most visibly in year-on-year inflation. **Combined with the stronger 30-day directional evidence, this suggests that the edge is largest early, remains material at ten days, and is still detectable immediately before release.**

The incremental information is stronger than the unconditional averages alone suggest. When Kalshi is surprised at one decimal, bottom-up identifies the correct direction in 71 of 98 cases (72.4%) and ML in 66 of 98 (67.3%). When the rounded model-market gap is at least 0.2 percentage point, the corresponding conditional rates are 12/12 (100%) for bottom-up and 17/19 (89%) for ML on the common ten-day panel. Across all available horizons, the large-gap directional signal is already visible at 30 days and remains present at 14 days and one hour. *(Note: These are point-forecast results; they should not be confused with the payoff of a particular Kalshi strike – see discussion below).*

1. Where the proprietary models are better

Bottom-up is the overall accuracy leader. Its pooled MAE is 8.6% below Consensus and 27.7% below Kalshi; its error dispersion is also the smallest. It wins three of the four series-level MAE comparisons—headline MoM, headline YoY and core YoY—while Consensus narrowly leads core MoM. Bottom-up is exact at one decimal in 46.3% of common-panel observations, compared with 42.9% for Consensus, 38.8% for ML and 33.3% for Kalshi. It also beats Kalshi's absolute error in 89 of 147 paired observations and beats Consensus in 74 of 147.

Table 1. Pooled ten-day performance on balanced panels from January 2023 onward.

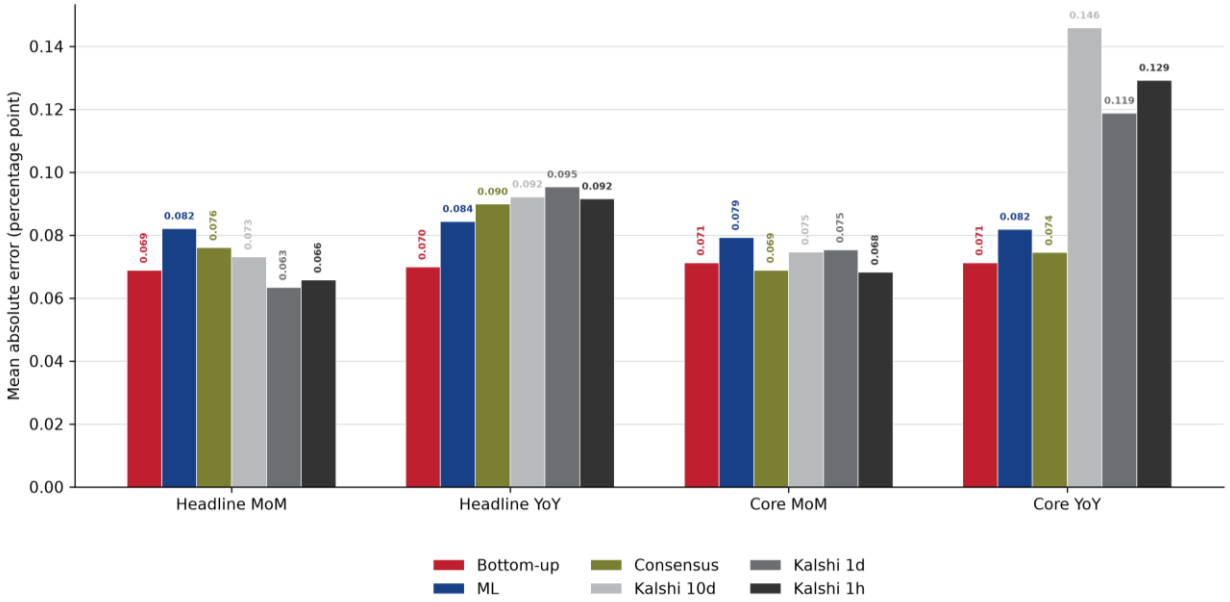
Model	N	MAE	RMSE	Error SD	Exact 0.1	Max miss
Bottom-up	147	**0.070**	**0.091**	**0.089**	**46.3%**	**0.250**
ML	147	0.082	0.106	0.106	38.8%	0.292

Model	N	MAE	RMSE	Error SD	Exact 0.1	Max miss
Consensus	147	0.077	0.102	0.100	42.9%	0.404
Kalshi 10d	147	0.097	0.143	0.143	33.3%	0.735

Source: authors' calculations. Pooled N sums the four series-specific common panels (Headline MoM 34, Headline YoY 35, Core MoM 39, Core YoY 39).

ML provides a complementary headline-year-on-year signal and materially improves on Kalshi overall. Its pooled MAE is 15.9% below Kalshi. In headline YoY, ML's MAE is 0.084, versus 0.090 for Consensus and 0.092 for Kalshi. For core YoY, the market benchmark is notably weak: Kalshi's MAE is 0.146, compared with 0.082 for ML and 0.071 for bottom-up.

Figure 1. Mean absolute error by CPI concept on balanced panels from January 2023.



Note: forecasts are compared only on dates available for all six series within each CPI concept. N is 34 for headline MoM, 35 for headline YoY, and 39 for each core concept. The y-axis is in percentage points.

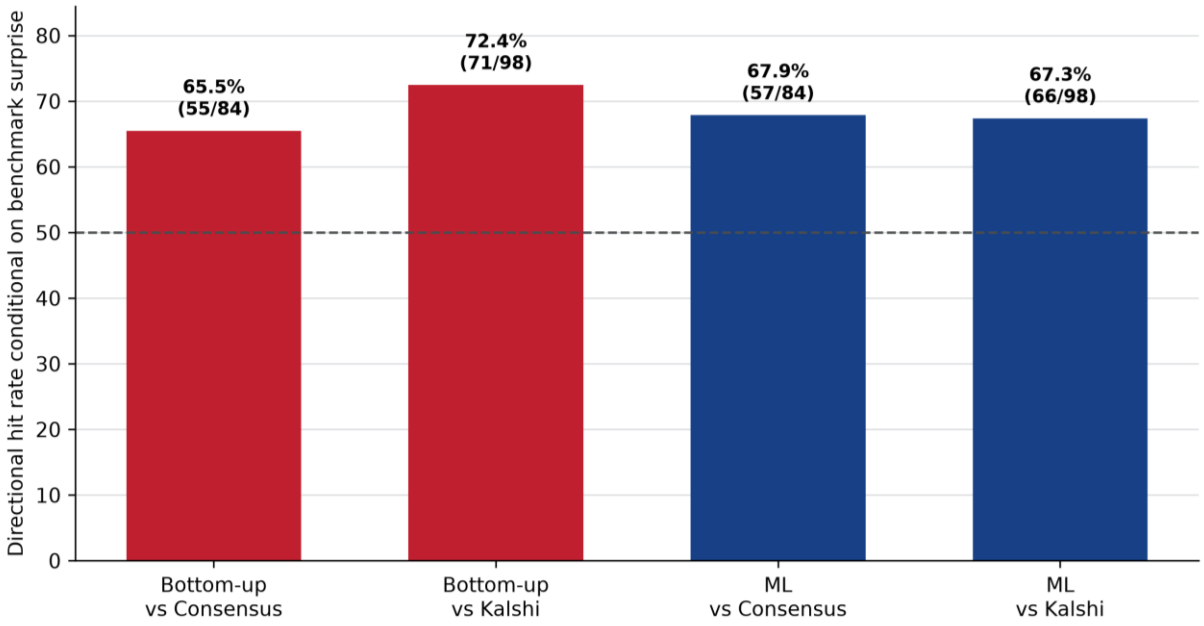
The market converges, but does not fully close the gap. Across the same 147 forecast-release pairs, Kalshi's pooled MAE improves from 0.097 at ten days to 0.089 at one day and 0.089 at one hour. At the series level, the one-day Kalshi forecast leads headline MoM with an MAE of 0.063, while the one-hour forecast marginally leads core MoM at 0.068. The more economically distinctive result is in the YoY rates: bottom-up records MAEs of 0.070 for headline YoY and 0.071 for core YoY, compared with Kalshi at 0.095 and 0.119 one day before release and 0.092 and 0.129 one hour before release. ML also remains more accurate than Kalshi in both YoY concepts. **Thus, the evidence is not that every proprietary forecast dominates every late market snapshot, but that the aggregate and especially YoY information advantage survives even after Kalshi has absorbed nearly all pre-release news.**

2. The models as directional risk overlays

Point-forecast accuracy is only the first use case. A portfolio manager often needs to know whether the benchmark is vulnerable to an upside or downside surprise. We therefore condition on Consensus or Kalshi being wrong at one decimal and ask whether the proprietary forecast was on the realized side of the benchmark. **Against Kalshi, bottom-up succeeds 72.4% of the time and ML 67.3%. Against Consensus, ML succeeds 67.9% and bottom-up 65.5%.**

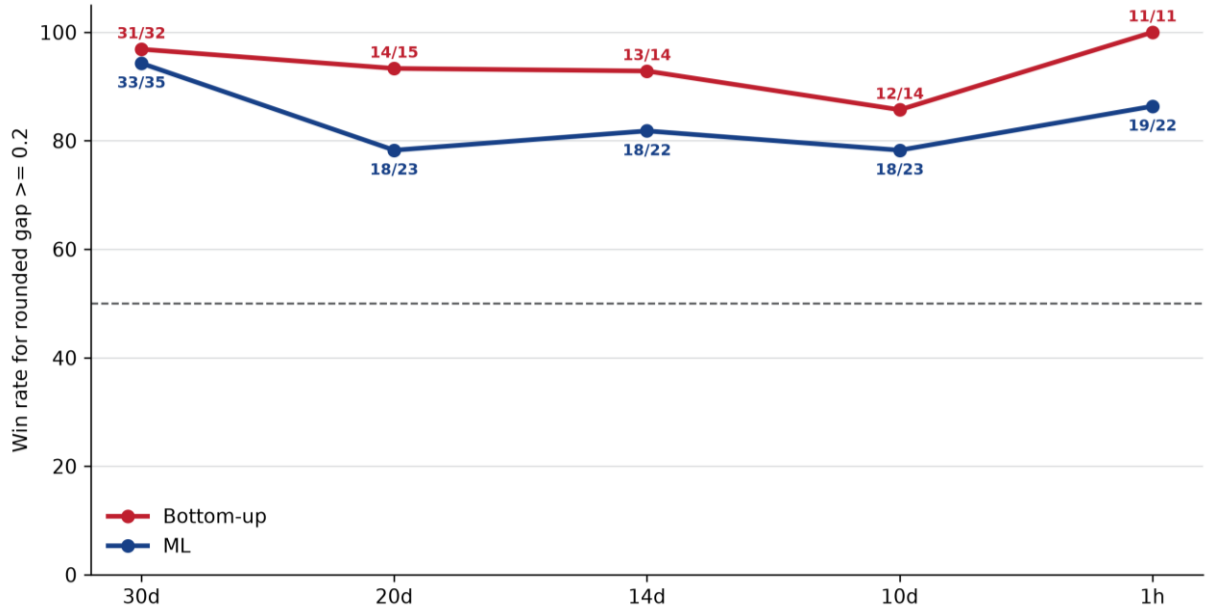
On the ten-day before the release common panel, when the model and Kalshi differ by at least 0.2 point after rounding and Kalshi is ultimately surprised, bottom-up is correct in 12 of 12 cases and ML in 17 of 19. Across all available horizons and CPI concepts, the 0.2-point signal produces 31/32 bottom-up wins and 33/35 ML wins at 30 days; 13/14 and 18/22 at 14 days; and 11/11 and 19/22 one hour before release.

Figure 2. Conditional directional hit rate when the benchmark is surprised at one decimal, pooled across headline/core and MoM/YoY on the common ten-day panel.



Note: A directional win means the release lies on the model's side of the benchmark; it does not require an exact forecast and does not by itself identify the best tradable strike.

Figure 3. High-conviction directional signal versus Kalshi. The plotted win-counts require a rounded model-market gap of at least 0.2 percentage point and pool all four CPI concepts.



Note: The sample uses all available post-July-2022 Kalshi observations by horizon. Directional accuracy is not the same as contract profitability because strikes, prices, spreads and fees matter.

3. From forecast disagreement to a tradable Kalshi position

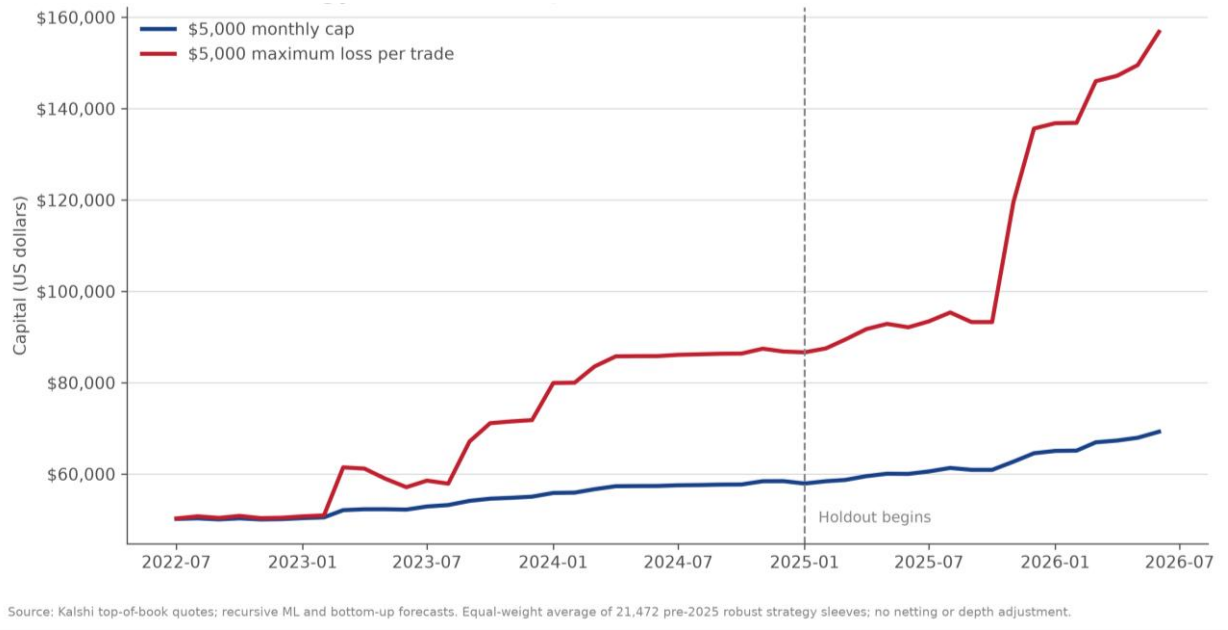
Our backtest permits YES and NO entries, but once a position is open it may only be held, closed, reduced or increased on the same side; it is never reversed. Entry horizons span 30 days through one hour, with all chronologically feasible later exits and settlement tested.

The strategy search evaluated 289,044 no-reversal variants. Discovery runs from July 2022 through December 2023, validation is calendar 2024, January 2025 through June 2026 is used as out-of-sample. A strategy enters the robust set only if it has at least eight pre-2025 positions, at least three positions in discovery and validation, positive P&L in both blocks, and positive pre-2025 P&L after removing its best month. Exactly 21,472 rules (7.4% of the universe) passed. The evidence suggests three practical rules. First, use bottom-up as the primary point-forecast risk overlay, especially for headline CPI and large disagreements with Kalshi. Second, use the ML model both as an independent headline-YoY view and, more importantly for execution, as a distributional engine: all 2,772 pre-2025-robust ML probability strategies are profitable in the subsequent holdout, with a median holdout P&L of $16,652 under the individual-strategy monthly budget. Third, combine the two models. The 50/50 average contributes 8,655 robust strategies, and 99.8% of its distribution rules remain profitable in holdout. Bottom-up remains strong in both forms: 99.96% of its 2,383 distribution strategies and 99.6% of its 4,482 robust point strategies remain profitable in holdout.

We evaluate two capital allocations: a 5,000$ monthly cap and a 5,000$ maximum loss per trade. Under the smoother $5,000 monthly-cap allocation, the equal-weight ensemble earns $19,277, ends at $69,277 and produces an 8.5% compound annual return with 3.2% volatility, a 2.58 zero-rate Sharpe ratio and a 0.9% maximum drawdown. It is profitable in 41 of 48 months, including $10,819 in

the holdout. The fixed-$5,000-maximum-loss-per-trade scaling ends at $156,832, a 213.7% cumulative return and 33.1% compound annual return, with 20.6% volatility, a 1.49 Sharpe ratio and a 7.1% maximum drawdown. Holdout P&L is $69,997. The monthly-cap curve is smoother; the fixed-risk curve has higher absolute and percentage return.

Figure 4. Capital evolution of the pre-2025 robust-strategy ensemble. Starting capital is $50,000; the dashed line marks the untouched holdout.



Note: The curve is the 1/N average of 21,472 independently feasible sleeves. Gross trade instances overlap heavily and are not unique market orders. Recorded top-of-book prices and Kalshi fees are used; order-book depth, slippage beyond the recorded spread and portfolio netting are unavailable.

What can be implemented. Freeze the signal definitions before the next release; compute bottom-up, ML and the 50/50 average at each permitted horizon; translate each point forecast into a recursively calibrated probability distribution; scan every live Kalshi threshold on both YES and NO; require positive net edge after ask, spread and fee; net contracts that express the same release risk; and cap total monthly loss. The monthly-cap ensemble is the more defensible starting point because it has the higher Sharpe ratio and lower drawdown. The fixed-per-trade curve is best interpreted as a capacity and scaling scenario rather than a recommendation to deploy that size immediately.

What must still be proved. The sample is short, large-gap cells can contain few events, the strategy search is broad and historical depth is unavailable. The next stage is a frozen prospective shadow book with archived quotes, order-book depth, decision timestamps and no parameter changes. The backtest supports the existence of an economically useful signal; it is not a live track record or a guarantee of future profits.

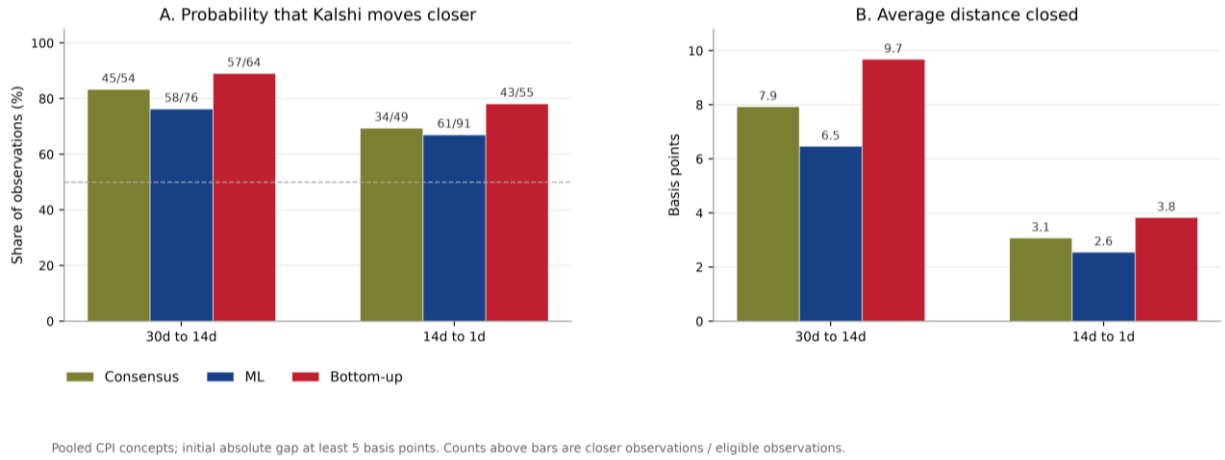
4. Trading Kalshi's convergence toward the proprietary forecasts

The preceding strategies ask whether a Kalshi contract is mispriced relative to the eventual CPI release. A separate exercise asks a different question: does Kalshi move toward an already-available forecast before settlement? This creates a potential mark-to-market trade that can be opened early

and closed before the release. For every CPI concept-month, we hold the benchmark forecast fixed and compare its absolute distance from Kalshi's reconstructed distribution mean at two horizons (14 days to 1 day before the release, and 30 days to 14 days before the release).

The strongest convergence occurs from 30 days to 14 days. Conditional on an initial discrepancy of at least 5 basis points, Kalshi moves closer to bottom-up in 57 of 64 observations (89.1%), to ML in 58 of 76 (76.3%) and to the reported Consensus in 45 of 54 (83.3%). The average distance closed is 9.7 basis points for bottom-up, 6.5 for ML and 7.9 for Consensus. Exact one-sided binomial p-values against a 50% null are below 0.000003 in all three cases. Convergence remains present from 14 days to one day, although the average movement is smaller: 3.8 basis points toward bottom-up, 2.6 toward ML and 3.1 toward Consensus.

Figure 5. Kalshi convergence toward fixed forecasts before release. Panel A reports the frequency of a decline in absolute distance; Panel B reports the mean distance closed. The sample pools headline and core CPI, month on month and year on year, and requires an initial gap of at least 5 basis points.



Note: July 2022-June 2026, subject to series-specific Kalshi coverage. Consensus is the reported survey forecast for the release, not a timestamped 30-day vintage; its convergence result is descriptive and is not used in the trading backtest.

Table 2. Kalshi convergence conditional on an initial gap of at least 5 basis points.

Window	Reference	N	Closer	Rate	Initial gap	Final gap	Gap closed
30d to 14d	Bottom-up	64	57/64	89.1%	15.8	6.1	9.7
30d to 14d	ML	76	58/76	76.3%	16.2	9.7	6.5
30d to 14d	Consensus	54	45/54	83.3%	14.4	6.5	7.9
14d to 1d	Bottom-up	55	43/55	78.2%	12.5	8.6	3.8
14d to 1d	ML	91	61/91	67.0%	11.9	9.3	2.6
14d to 1d	Consensus	49	34/49	69.4%	11.1	8.0	3.1

Note: Initial gap, final gap and gap closed are means in basis points. At a 10-basis-point initial-gap threshold, the 30d-to-14d convergence rates rise to 97.4% for bottom-up, 87.2% for ML and 96.4% for Consensus; the corresponding 14d-to-1d rates are 80.0%, 70.3% and 86.7%.

A directly executable point-convergence rule. At the 30-day snapshot, calculate the 50/50 average of the bottom-up and ML point forecasts for each of the four CPI concepts. Trade only when its distance from Kalshi's distribution mean is at least 5 basis points. If the model is above Kalshi, buy YES on the inclusive threshold one rung below the rounded market mean; if it is below Kalshi, buy NO on the threshold at the rounded market mean. Rank signals by absolute gap, take at most the two strongest in the month, reject entry prices above $0.90, split a $5,000 monthly budget equally across accepted positions, buy at the recorded ask, and sell the entire position at the recorded 14-day bid. Kalshi taker fees are charged on both legs and quotes more than 24 hours stale are excluded. The rule never reverses a position.

Table 3. Point-convergence strategy using executable entry and exit quotes.

Window	Signal	Positions	Wins	Win rate	P&L	ROI	2025+ wins	2025+ P&L	2025+ ROI
30d to 14d	Bottom-up	33	24/33	72.7%	$11,120	13.8%	18/22	$9,702	20.1%
30d to 14d	ML	36	26/36	72.2%	$8,748	11.3%	17/24	$6,710	14.3%
30d to 14d	50/50 average	28	23/28	82.1%	$12,072	18.0%	17/20	$9,006	19.5%
14d to 1d	Bottom-up	35	19/35	54.3%	$2,454	2.8%	10/14	$1,414	4.4%
14d to 1d	ML	54	22/54	40.7%	$-7,769	-6.0%	13/26	$-1,444	-2.6%
14d to 1d	50/50 average	37	17/37	45.9%	$-2,281	-2.2%	11/19	$809	1.7%

Note: Full sample: July 2022-June 2026. The untouched reporting block is January 2025-June 2026. ROI is P&L divided by capital deployed, not return on the $50,000 research capital base.

The combined proprietary signal is the preferred specification. The 30d-to-14d 50/50 rule executes 28 positions, wins 23 (82.1%), earns $12,072 and delivers an 18.0% return on deployed capital. In the January 2025-June 2026 reporting block it wins 17 of 20 positions, earns $9,006 and returns 19.5%. Bottom-up and ML also earn positive full-sample and post-2024 profits separately. By contrast, mechanically carrying the same rule from 14 days to one day is negative over the full sample.

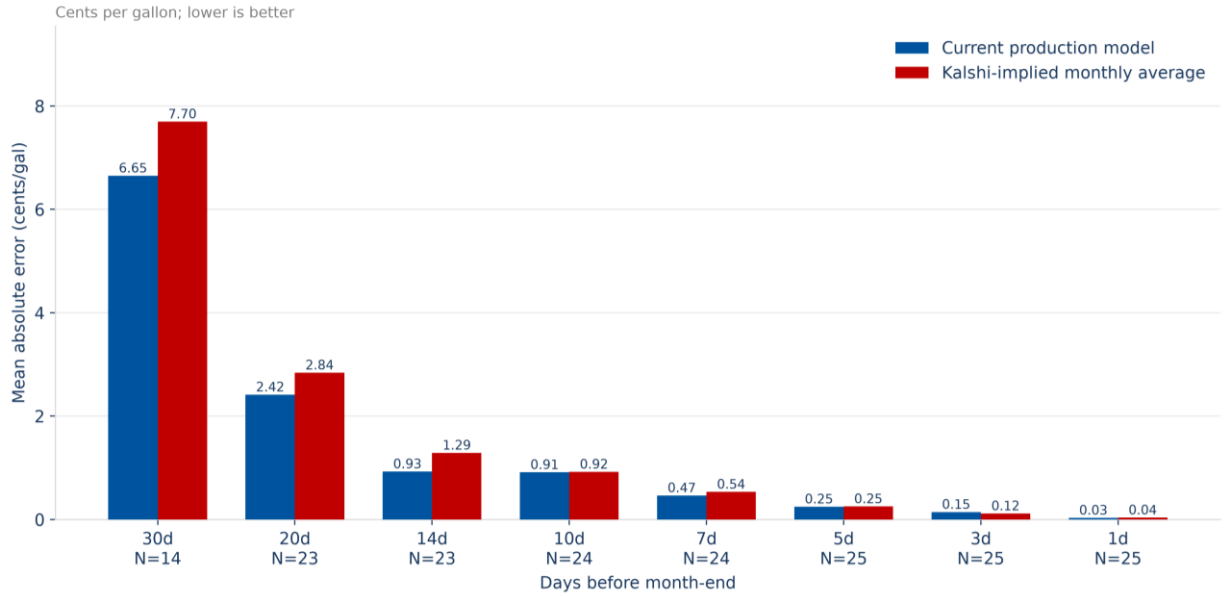
5. Gasoline forecasting and executable Kalshi strategies

The gasoline framework is a single forecasting model. The model uses the national AAA regular-gasoline series together with the oil information then available. It estimates the calendar-month average for the accuracy comparison and the relevant terminal AAA price for the tradable next-day, next-Monday and month-end Kalshi contracts (the model also forecasts the average of the month which feeds into our bottom up and ML model forecasts). The model point forecast is combined with recursively estimated, horizon-matched historical errors to obtain a full predictive distribution. Every historical forecast is reconstructed using only the information available at its origin, and every Kalshi quote observed after the relevant cutoff is excluded.

The first test asks whether the production nowcast contains incremental information relative to Kalshi. Because the monthly Kalshi contract settles the terminal AAA price rather than the calendar-month average, we reconstruct the market-implied terminal distribution from the full strike ladder and combine each possible terminal outcome with realized month-to-date gasoline prices through a linear bridge. This produces a synthetic Kalshi-implied monthly-average benchmark on a strictly matched real-time sample. **Our model records the lower mean absolute error at seven of eight horizons. Its MAE is 6.65 cents per gallon at 30 days, compared with 7.70 cents for the Kalshi proxy; 2.42 versus 2.84 cents at 20 days; and 0.93 versus 1.29 cents at 14 days. Kalshi is better only at three days, by approximately 0.03 cent, while the production model is again marginally better at one day.**

The result survives a stricter common-sample test. Restricting every horizon to the same 14 events leaves the production model ahead at seven of eight horizons, with a 33.4% lower MAE at 14 days and a 17.9% lower MAE at 10 days. The pattern is economically coherent and falsifiable. The advantage is broad early in the month, is largest near the 14-day horizon, narrows as Kalshi absorbs the incoming daily data, and is almost exhausted near settlement. This timing identifies where the model should be most valuable, while the direct trading test below determines whether the advantage survives strike selection, spreads and fees.

Figure 6. Matched real-time mean absolute error for the calendar-month average of regular gasoline. Lower bars indicate greater accuracy; N is reported for every horizon.



Source: authors' calculations using AAA regular-gasoline data, real-time oil vintages and strict pre-cutoff Kalshi snapshots. MAE is in cents per gallon. Because the Kalshi contract settles the terminal AAA price, the market series shown here is a bridge-implied analytical benchmark rather than a directly tradable monthly-average contract.

Table 4. Matched gasoline MAE and relative advantage by forecast horizon.

Horizon	N	Production MAE	Kalshi-proxy MAE	Lower MAE	Relative gap
30d	14	**6.651**	7.699	**Production model**	13.6%
20d	23	**2.417**	2.841	**Production model**	14.9%
14d	23	**0.928**	1.291	**Production model**	28.1%
10d	24	**0.914**	0.922	**Production model**	0.9%
7d	24	**0.466**	0.535	**Production model**	13.0%
5d	25	**0.249**	0.253	**Production model**	1.8%
3d	25	0.146	**0.116**	**Kalshi proxy**	25.8%
1d	25	**0.033**	0.037	**Production model**	11.8%

Relative gap is the absolute percentage difference between the two MAEs. At one day, both errors are extremely small because most days entering the monthly average have already been observed. Historical results are research simulations, not a prospective track record.

The accuracy comparison establishes the information edge, while contract-level expected value determines whether it can be traded. At each decision time, the horizon-specific predictive distribution is evaluated against every live Kalshi strike on both the YES and NO sides. A position qualifies only when the model probability exceeds the executable ask, the Kalshi taker fee and the required five-percentage-point margin. Point-gap rules and paired vertical structures are admitted only when their exact historical payoff clears the same execution test. Because neighboring strikes and repeated entries share settlement risk, the production process ranks qualifying contracts by net edge and caps funded loss rather than treating every signal as independent.

The model is evaluated every day, but a trade is placed only when a historically validated condition is met. The research curves use US$10,000 of starting capital and a US$1,000 maximum loss per position, recorded executable asks and Kalshi fees. Positions are held to settlement unless the rule defines a paired payoff.

This architecture separates forecast quality from trade construction. The point forecast determines the direction and size of a disagreement, while the recursively calibrated distribution prices the probability of each strike and the joint payoff of vertical combinations. It therefore allows the strategy to buy either YES or NO, including tail contracts that are attractive even when the market median is close to the model forecast. The backtest reruns the production model at every historical decision time and rejects any quote unavailable at that time.

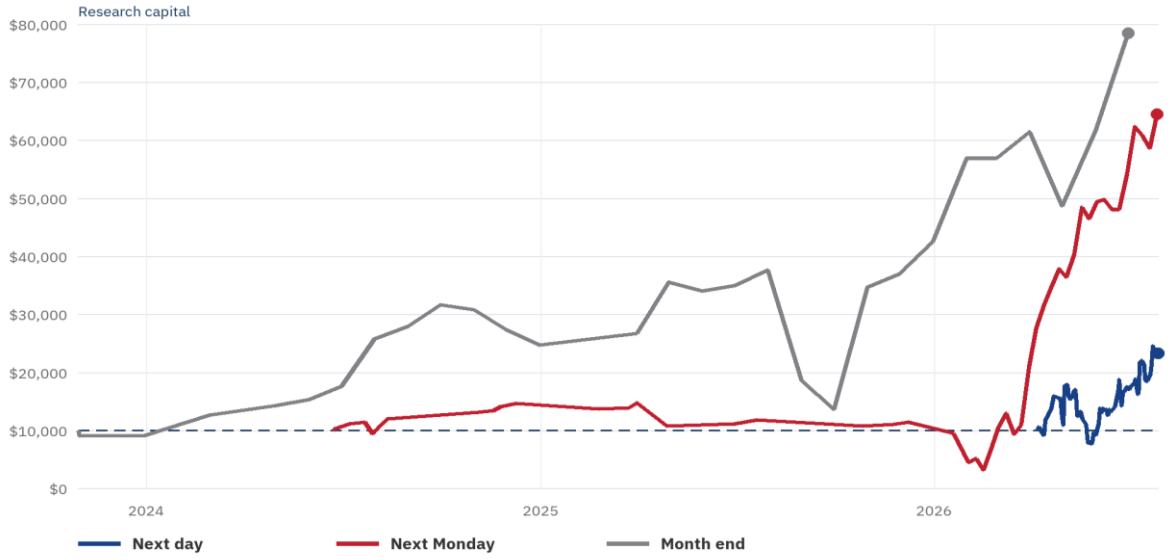
The strongest evidence comes from the next-Monday conviction ladder. The contract is reevaluated every day from seven days to one day before settlement. The rule buys one US$1,000 position when the model and Kalshi median differ by at least two cents, adds a second position at three cents and a third at five cents, and rejects contracts costing more than $0.80. Across 697 executable decision points, the ladder placed 208 positions in 46 settlement events, won 74.0%, earned $54,522 and returned 26.2% on deployed capital. Both chronological halves are profitable, the event-clustered random-sign probability is 1.6%, and the neighboring two-cent, three-cent and five-cent rules are each profitable in both halves.

Daily evaluation does not imply compulsory daily turnover. Forcing the highest-scored next-Monday position on every executable decision produced 661 positions and lost $142,408 after fees under the current US$1,000 risk scale. The profitable ladder traded on 15.6% of executable weekly decision points. The month-end core stack traded on 32.1%, while the next-day rules generated a signal on 68.5% of their executable dates. Selectivity is therefore part of the estimated edge.

The month-end implementation combines two independent sleeves. The full-distribution sleeve opens the first contract in each month whose estimated probability exceeds the executable price and fee by at least five percentage points. The point sleeve buys the model side of the Kalshi median on every eligible day once the gap reaches three cents. Together they generated 255 positions across 28 settlement months, won 67.5%, earned $68,527 and returned 26.9% on deployed capital. The profitable-month rate is 71.4%, both chronological halves are positive, and maximum drawdown is $25,990. Repeated entries within a month share one settlement outcome, so the position count overstates statistical independence.

The next-day stack generated 147 positions over 76 daily events, won 61.9%, earned $13,259 and returned 9.0% on deployed capital. Its interval component buys YES at the lower strike and NO at the higher strike, paying $1 outside the interval and $2 inside it. That component alone placed 72 positions, won 65.3%, earned $4,667 and limited maximum drawdown to $983. The available next-day history begins in April 2026 and the combined event-clustered random-sign probability is 33.8%, so the result is operationally useful but carries less evidentiary weight than the longer weekly and month-end panels.

Figure 7. Historical research capital for the preferred executable gasoline strategies under the current production risk scale.



Source: authors' calculations using real-time production-model forecasts, recorded executable Kalshi prices and fees. Each curve starts with US$10,000 and applies a US$1,000 maximum loss per position. The next-day curve combines all qualifying point and vertical signals; the next-Monday curve is the two-cent, three-cent and five-cent conviction ladder; the month-end curve combines the three-cent daily point sleeve with the first qualifying probability signal.

Table 5. Executable gasoline-strategy results under the current US$1,000 risk scale and daily decision coverage.

Strategy	Positions	Events	Win rate	Net P&L	ROI	Max DD	Signal coverage	Cluster p
Next day all signals	147	76	61.9%	$13,259	9.0%	$10,236	68.5%	0.338
Next Monday ladder	208	46	74.0%	$54,522	26.2%	$11,582	15.6%	0.016
Month end core stack	255	28	67.5%	$68,527	26.9%	$25,990	32.1%	0.124

Note: The combined evidence supports a differentiated conclusion. The weekly ladder has the strongest event-level statistical support and remains profitable across neighboring thresholds and chronological halves. The month-end stack produces the largest net profit and exploits both point and distributional information, although its repeated entries create material within-month concentration. The next-day stack broadens daily coverage and demonstrates that tail and interval structures can add value, while its short sample requires prospective confirmation. Signal coverage is the share of executable decision points on which a rule traded, and cluster p is a two-sided random-sign test on event-level P&L without adjustment for the strategy search. These are historical research simulations rather than a live track record.

Appendix A: complete statistical tables

The appendix separates strict common-panel comparisons from available-observation diagnostics. This prevents a model with more history from being rewarded or penalized simply because it is scored in different months. Bold shaded cells identify the best model within each CPI concept for the stated metric.

Table A1. Ten-day accuracy on the strict four-model common panel.

Series	Model	N	MAE	RMSE	Error SD	Exact 0.1	Max raw
Headline MoM	Bottom-up	34	**0.069**	**0.089**	**0.088**	**47.1%**	**0.240**
Headline MoM	ML	34	0.082	0.104	0.105	41.2%	0.256
Headline MoM	Consensus	34	0.076	0.098	0.098	44.1%	0.320
Headline MoM	Kalshi 10d	34	0.073	0.092	0.092	41.2%	0.266
Headline YoY	Bottom-up	35	**0.070**	**0.090**	**0.088**	**48.6%**	**0.240**
Headline YoY	ML	35	0.084	0.106	0.108	31.4%	0.266
Headline YoY	Consensus	35	0.090	0.124	0.123	42.9%	0.404
Headline YoY	Kalshi 10d	35	0.092	0.128	0.125	34.3%	0.372
Core MoM	Bottom-up	39	0.071	0.092	0.091	**46.2%**	0.250
Core MoM	ML	39	0.079	0.105	0.107	43.6%	0.284
Core MoM	Consensus	39	**0.069**	**0.090**	**0.089**	43.6%	0.243
Core MoM	Kalshi 10d	39	0.075	0.093	0.092	35.9%	**0.200**
Core YoY	Bottom-up	39	**0.071**	**0.092**	**0.091**	**43.6%**	0.250
Core YoY	ML	39	0.082	0.109	0.110	38.5%	0.292
Core YoY	Consensus	39	0.074	0.093	0.094	41.0%	**0.234**
Core YoY	Kalshi 10d	39	0.146	0.215	0.214	23.1%	0.735

Each series uses only months in which bottom-up, ML, Consensus and Kalshi 10d are all observed. The four pooled N values sum to 147.

Table A2. Available-observation diagnostics from January 2023 onward; N is always reported.

Series	Model	N	MAE	RMSE	Error SD	Exact 0.1	Max raw
Headline MoM	Bottom-up	40	**0.071**	0.092	**0.089**	**47.5%**	**0.240**
Headline MoM	ML	41	0.080	0.102	0.103	39.0%	0.256
Headline MoM	Consensus	35	0.076	0.098	0.097	42.9%	0.320
Headline MoM	Kalshi 10d	40	0.072	**0.091**	0.091	45.0%	0.266
Headline YoY	Bottom-up	40	**0.071**	**0.092**	**0.089**	**50.0%**	**0.240**
Headline YoY	ML	41	0.083	0.105	0.106	31.7%	0.266
Headline YoY	Consensus	36	0.090	0.123	0.121	41.7%	0.404
Headline YoY	Kalshi 10d	40	0.098	0.136	0.135	32.5%	0.372
Core MoM	Bottom-up	40	0.071	0.092	0.090	**47.5%**	0.250
Core MoM	ML	41	0.078	0.103	0.105	43.9%	0.284
Core MoM	Consensus	40	**0.069**	**0.090**	**0.088**	42.5%	0.243

Series	Model	N	MAE	RMSE	Error SD	Exact 0.1	Max raw
Core MoM	Kalshi 10d	40	0.074	0.092	0.092	37.5%	**0.200**
Core YoY	Bottom-up	40	**0.071**	**0.092**	**0.090**	**42.5%**	**0.250**
Core YoY	ML	41	0.080	0.107	0.108	39.0%	0.292
Core YoY	Consensus	41	0.083	0.111	0.110	39.0%	0.401
Core YoY	Kalshi 10d	39	0.146	0.215	0.214	23.1%	0.735

Available N ranges from 35 to 41. These cells are useful for model diagnostics but are not the primary ranking because the models can be scored in different months.

Table A3. Maximum misses on the strict common panel.

Series	Model	N	Max raw	Displayed gap	Month	Forecast	Release
Headline MoM	Bottom-up	34	0.240	0.2	Jun 2026	-0.180	-0.420
Headline MoM	ML	34	0.256	0.3	Jun 2026	-0.164	-0.420
Headline MoM	Consensus	34	0.320	0.3	Jun 2026	-0.100	-0.420
Headline MoM	Kalshi 10d	34	0.266	0.2	Jun 2026	-0.154	-0.420
Headline YoY	Bottom-up	35	0.240	0.2	Jun 2026	3.704	3.464
Headline YoY	ML	35	0.266	0.3	Jun 2026	3.729	3.464
Headline YoY	Consensus	35	0.404	0.4	Nov 2025	3.100	2.696
Headline YoY	Kalshi 10d	35	0.372	0.3	Jun 2024	3.342	2.970
Core MoM	Bottom-up	39	0.250	0.2	Jun 2026	0.230	-0.020
Core MoM	ML	39	0.284	0.2	Jan 2025	0.162	0.446
Core MoM	Consensus	39	0.243	0.2	Mar 2025	0.300	0.057
Core MoM	Kalshi 10d	39	0.200	0.2	Jun 2026	0.180	-0.020
Core YoY	Bottom-up	39	0.250	0.2	Jun 2026	2.816	2.566
Core YoY	ML	39	0.292	0.3	Jan 2025	2.991	3.283
Core YoY	Consensus	39	0.234	0.2	Jun 2026	2.800	2.566
Core YoY	Kalshi 10d	39	0.735	0.7	May 2023	4.603	5.338

Max raw is the largest unrounded absolute error. Displayed gap is the absolute difference between forecast and release after each is rounded to one decimal; it is not the rounded raw error.

Table A4. Pairwise absolute-error wins on the common panel.

Panel	Model	Benchmark	N	Model wins	Benchmark wins	Win rate
Headline pooled	ML	Consensus	69	34	35	49.3%
Headline pooled	ML	Kalshi 10d	69	29	40	42.0%
Headline pooled	Bottom-up	Consensus	69	40	28	58.8%
Headline pooled	Bottom-up	Kalshi 10d	69	38	31	55.1%
Core pooled	ML	Consensus	78	34	44	43.6%
Core pooled	ML	Kalshi 10d	78	42	36	53.8%
Core pooled	Bottom-up	Consensus	78	34	38	47.2%
Core pooled	Bottom-up	Kalshi 10d	78	51	27	65.4%

Panel	Model	Benchmark	N	Model wins	Benchmark wins	Win rate
All four pooled	ML	Consensus	147	68	79	46.3%
All four pooled	ML	Kalshi 10d	147	71	76	48.3%
All four pooled	Bottom-up	Consensus	147	74	66	52.9%
All four pooled	Bottom-up	Kalshi 10d	147	89	58	60.5%

Model wins require lower unrounded error; ties remain in N but not the win counts or rates.

Table A5. Conditional direction when Consensus or Kalshi is surprised at one decimal.

Panel	Model	Benchmark	Surprises	Directional wins	Hit rate	Gap >=0.2
Headline pooled	ML	Consensus	39	30/39	76.9%	2/3 | 66.7%
Headline pooled	ML	Kalshi 10d	43	29/43	67.4%	8/9 | 88.9%
Headline pooled	Bottom-up	Consensus	39	28/39	71.8%	1/1 | 100.0%
Headline pooled	Bottom-up	Kalshi 10d	43	27/43	62.8%	4/4 | 100.0%
Core pooled	ML	Consensus	45	27/45	60.0%	0/0
Core pooled	ML	Kalshi 10d	55	37/55	67.3%	9/10 | 90.0%
Core pooled	Bottom-up	Consensus	45	27/45	60.0%	0/0
Core pooled	Bottom-up	Kalshi 10d	55	44/55	80.0%	8/8 | 100.0%
All four pooled	ML	Consensus	84	57/84	67.9%	2/3 | 66.7%
All four pooled	ML	Kalshi 10d	98	66/98	67.3%	17/19 | 89.5%
All four pooled	Bottom-up	Consensus	84	55/84	65.5%	1/1 | 100.0%
All four pooled	Bottom-up	Kalshi 10d	98	71/98	72.4%	12/12 | 100.0%

The final column applies the additional rounded model-benchmark gap of at least 0.2 percentage point within benchmark-surprise observations.

Table A6. Large-gap directional signal versus Kalshi across horizons, all four CPI concepts pooled.

Horizon	Model	Available N	Signals	Wins	Win rate
30d	Bottom-up	137	32	31	96.9%
30d	ML	137	35	33	94.3%
20d	Bottom-up	156	15	14	93.3%
20d	ML	156	23	18	78.3%
14d	Bottom-up	157	14	13	92.9%
14d	ML	157	22	18	81.8%
10d	Bottom-up	159	14	12	85.7%
10d	ML	159	23	18	78.3%
1h	Bottom-up	159	11	11	100.0%
1h	ML	159	22	19	86.4%

A signal requires a rounded model-Kalshi gap of at least 0.2. Available N varies with historical Kalshi coverage. The row is directional evidence, not a simulated contract payoff.

Table A7. Corrected trading-ensemble performance.

Sizing	End capital	Total	CAGR	Ann. vol.	Sharpe	Max DD	Holdout P&L
$5,000 monthly cap	$69,277	38.6%	8.5%	3.2%	2.58	-0.9%	$10,819
$5,000 maximum loss per trade	$156,832	213.7%	33.1%	20.6%	1.49	-7.1%	$69,997

Sharpe assumes a zero risk-free rate and monthly returns. Holdout is January 2025–June 2026. Starting capital is $50,000.

Table A8. Composition and holdout behavior of the pre-2025 robust set.

Model	Signal	Robust rules	Median train P&L	Median holdout P&L	Positive holdout	Median ROI
Bottom-up	Point	4482	$5,649	$10,619	99.6%	19.4%
Bottom-up	Distribution	2383	$6,268	$20,372	100.0%	37.6%
ML	Point	3180	$8,181	$-1,048	39.6%	7.9%
ML	Distribution	2772	$8,834	$16,652	100.0%	41.1%
50/50 average	Point	3534	$9,385	$1,888	76.1%	15.8%
50/50 average	Distribution	5121	$8,746	$16,652	99.8%	32.2%

The robust gate uses pre-2025 data only. Median P&L refers to individual $5,000-monthly-cap strategies before 1/N ensemble normalization. Average50 is the equal-weight forecast average.

Methodological notes and safeguards

Forecast comparison. The common-panel tables are the ranking tables. Available-observation tables are diagnostic. Consensus is observed only at one decimal; Kalshi is reconstructed from the relevant contract distribution. All horizon labels refer to distance from the release, with a 24-hour maximum quote-staleness filter.

Trading simulation. Every purchase uses the recorded ask, every sale the recorded bid, and Kalshi taker fees are charged on every leg. A settlement fallback is used when no valid exit bid exists. The simulation assumes the displayed top-of-book price can absorb the required quantity. No position reversal is permitted. Overlapping rules are not netted, so the ensemble is a research portfolio of strategy sleeves rather than an execution-ready list of orders.

Interpretation. The evidence is strongest when viewed as triangulation: bottom-up leads the common-panel point-forecast ranking; ML adds an independent headline-YoY signal and a strong probability-mapping engine; both anticipate the direction of Kalshi misses; and profitability persists in an untouched 2025–June 2026 holdout. The remaining step is prospective execution with frozen rules, archived depth and explicit portfolio-level netting.

Appendix B: Monthly forecast ledger - 10d

Each table reports the published release and every forecast used in the 10d (10 days before the release) comparison. A shaded, bold forecast is a row winner: it has the smallest absolute miss after both the forecast and the release are rounded to one decimal. Ties are retained. The final column names every winner explicitly.

Headline MoM - monthly forecasts (N=34).

Reference month	Published	Bottom-up	ML	Consensus	Kalshi 10d	Winner(s)
Jun 2023	0.180	0.260	**0.239**	0.300	0.255	**ML**
Jul 2023	0.167	**0.160**	**0.243**	**0.200**	**0.213**	**Bottom-up; ML; Consensus; Kalshi 10d**
Aug 2023	0.631	**0.640**	**0.555**	**0.600**	**0.641**	**Bottom-up; ML; Consensus; Kalshi 10d**
Sep 2023	0.396	0.240	0.325	0.300	**0.358**	**Kalshi 10d**
Oct 2023	0.045	**0.140**	**0.067**	**0.100**	**0.105**	**Bottom-up; ML; Consensus; Kalshi 10d**
Nov 2023	0.097	**0.110**	**0.146**	**0.100**	**0.065**	**Bottom-up; ML; Consensus; Kalshi 10d**
Dec 2023	0.303	**0.250**	0.203	0.200	0.237	**Bottom-up**
Jan 2024	0.305	**0.150**	0.115	**0.200**	**0.186**	**Bottom-up; Consensus; Kalshi 10d**
Feb 2024	0.442	**0.420**	0.347	**0.400**	0.458	**Bottom-up; Consensus**
Mar 2024	0.378	**0.350**	**0.365**	0.300	0.267	**Bottom-up; ML**
Apr 2024	0.313	0.390	**0.267**	0.400	0.382	**ML**
May 2024	0.006	**0.140**	**0.133**	**0.100**	**0.139**	**Bottom-up; ML; Consensus; Kalshi 10d**
Jun 2024	-0.056	0.090	**0.015**	0.100	0.060	**ML**
Jul 2024	0.155	**0.210**	**0.185**	**0.200**	**0.162**	**Bottom-up; ML; Consensus; Kalshi 10d**
Aug 2024	0.187	**0.180**	**0.185**	**0.200**	**0.172**	**Bottom-up; ML; Consensus; Kalshi 10d**
Sep 2024	0.180	**0.090**	**0.118**	**0.100**	**0.072**	**Bottom-up; ML; Consensus; Kalshi 10d**
Oct 2024	0.244	0.250	**0.216**	**0.200**	0.409	**ML; Consensus**
Nov 2024	0.313	**0.300**	0.242	**0.300**	**0.254**	**Bottom-up; Consensus; Kalshi 10d**
Dec 2024	0.393	**0.350**	**0.364**	0.300	**0.382**	**Bottom-up; ML; Kalshi 10d**
Jan 2025	0.467	**0.350**	0.225	0.300	0.306	**Bottom-up**
Feb 2025	0.216	**0.250**	**0.279**	**0.300**	**0.321**	**Bottom-up; ML; Consensus; Kalshi 10d**
Mar 2025	-0.050	**0.130**	0.172	**0.100**	**0.101**	**Bottom-up; Consensus; Kalshi 10d**
Apr 2025	0.221	**0.310**	**0.147**	**0.300**	**0.255**	**Bottom-up; ML; Consensus; Kalshi 10d**
May 2025	0.081	0.190	**0.110**	0.200	0.179	**ML**
Jun 2025	0.287	0.230	0.218	**0.300**	0.241	**Consensus**
Jul 2025	0.197	**0.220**	**0.208**	**0.200**	0.260	**Bottom-up; ML; Consensus**
Aug 2025	0.382	**0.320**	**0.324**	**0.300**	**0.328**	**Bottom-up; ML; Consensus; Kalshi 10d**
Dec 2025	0.307	0.370	**0.251**	**0.300**	**0.264**	**ML; Consensus; Kalshi 10d**
Jan 2026	0.171	**0.220**	**0.193**	0.300	**0.229**	**Bottom-up; ML; Kalshi 10d**
Feb 2026	0.267	**0.270**	0.354	**0.300**	**0.269**	**Bottom-up; Consensus; Kalshi 10d**
Mar 2026	0.865	**0.910**	0.976	**0.900**	**0.885**	**Bottom-up; Consensus; Kalshi 10d**
Apr 2026	0.640	**0.610**	0.522	**0.600**	**0.562**	**Bottom-up; Consensus; Kalshi 10d**
May 2026	0.470	**0.520**	0.626	**0.500**	**0.518**	**Bottom-up; Consensus; Kalshi 10d**
Jun 2026	-0.420	**-0.180**	**-0.164**	-0.100	**-0.154**	**Bottom-up; ML; Kalshi 10d**

Values are percentage points and are displayed to three decimals. Winner determination uses one-decimal displayed values, not the extra digits shown for auditability.

Headline YoY - monthly forecasts (N=35).

Reference month	Published	Bottom-up	ML	Consensus	Kalshi 10d	Winner(s)
Jun 2023	3.071	3.150	**3.131**	**3.100**	3.045	**ML; Consensus**
Jul 2023	3.288	**3.281**	3.367	**3.300**	3.234	**Bottom-up; Consensus**
Aug 2023	3.723	**3.731**	3.644	3.600	**3.677**	**Bottom-up; Kalshi 10d**
Sep 2023	3.687	3.531	3.614	3.600	**3.658**	**Kalshi 10d**
Oct 2023	3.251	**3.346**	**3.274**	**3.300**	3.434	**Bottom-up; ML; Consensus**
Nov 2023	3.133	**3.146**	3.184	**3.100**	**3.111**	**Bottom-up; Consensus; Kalshi 10d**
Dec 2023	3.316	**3.263**	3.213	3.200	**3.261**	**Bottom-up; Kalshi 10d**
Jan 2024	3.088	2.933	2.892	2.900	**2.989**	**Kalshi 10d**
Feb 2024	3.157	**3.135**	**3.060**	**3.100**	**3.121**	**Bottom-up; ML; Consensus; Kalshi 10d**
Mar 2024	3.487	**3.459**	**3.474**	3.400	3.338	**Bottom-up; ML**
Apr 2024	3.361	**3.438**	3.314	**3.400**	3.478	**Bottom-up; Consensus**
May 2024	3.244	**3.379**	**3.376**	**3.400**	**3.401**	**Bottom-up; ML; Consensus; Kalshi 10d**
Jun 2024	2.970	3.116	**3.043**	3.100	3.342	**ML**
Jul 2024	2.941	2.997	2.973	3.000	**2.930**	**Kalshi 10d**
Aug 2024	2.607	**2.600**	**2.605**	**2.600**	2.518	**Bottom-up; ML; Consensus**
Sep 2024	2.426	2.337	**2.363**	2.300	2.308	**ML**
Oct 2024	2.579	**2.585**	**2.551**	**2.600**	**2.566**	**Bottom-up; ML; Consensus; Kalshi 10d**
Nov 2024	2.719	**2.707**	2.647	**2.700**	**2.728**	**Bottom-up; Consensus; Kalshi 10d**
Dec 2024	2.871	2.828	2.841	**2.900**	**2.871**	**Consensus; Kalshi 10d**
Jan 2025	2.991	**2.874**	2.743	**2.900**	**2.929**	**Bottom-up; Consensus; Kalshi 10d**
Feb 2025	2.802	**2.836**	2.866	2.900	2.895	**Bottom-up**
Mar 2025	2.382	2.562	2.609	2.600	**2.544**	**Kalshi 10d**
Apr 2025	2.325	2.414	**2.250**	2.400	2.406	**ML**
May 2025	2.377	2.486	**2.407**	2.500	**2.444**	**ML; Kalshi 10d**
Jun 2025	2.680	2.623	2.610	**2.700**	2.626	**Consensus**
Jul 2025	2.743	**2.766**	**2.754**	**2.800**	**2.791**	**Bottom-up; ML; Consensus; Kalshi 10d**
Aug 2025	2.939	**2.876**	**2.878**	**2.900**	**2.873**	**Bottom-up; ML; Consensus; Kalshi 10d**
Nov 2025	2.696	**2.802**	**2.778**	3.100	3.014	**Bottom-up; ML**
Dec 2025	2.653	**2.716**	2.596	**2.700**	**2.681**	**Bottom-up; Consensus; Kalshi 10d**
Jan 2026	2.391	**2.440**	**2.414**	2.500	2.457	**Bottom-up; ML**
Feb 2026	2.434	**2.437**	2.523	**2.400**	2.466	**Bottom-up; Consensus**
Mar 2026	3.286	**3.331**	3.399	**3.300**	**3.336**	**Bottom-up; Consensus; Kalshi 10d**
Apr 2026	3.779	**3.749**	**3.657**	**3.700**	**3.688**	**Bottom-up; ML; Consensus; Kalshi 10d**
May 2026	4.167	**4.217**	4.329	**4.200**	4.253	**Bottom-up; Consensus**
Jun 2026	3.464	**3.704**	**3.729**	3.800	3.800	**Bottom-up; ML**

Values are percentage points and are displayed to three decimals. Winner determination uses one-decimal displayed values, not the extra digits shown for auditability.

Core MoM - monthly forecasts (N=39).

Reference month	Published	Bottom-up	ML	Consensus	Kalshi 10d	Winner(s)
Jan 2023	0.412	**0.360**	**0.396**	**0.400**	**0.361**	**Bottom-up; ML; Consensus; Kalshi 10d**
Feb 2023	0.452	**0.400**	**0.408**	**0.400**	**0.401**	**Bottom-up; ML; Consensus; Kalshi 10d**
Mar 2023	0.385	**0.440**	**0.358**	**0.400**	0.456	**Bottom-up; ML; Consensus**
Apr 2023	0.409	0.480	0.256	**0.400**	**0.406**	**Consensus; Kalshi 10d**
May 2023	0.436	**0.370**	0.330	**0.400**	0.301	**Bottom-up; Consensus**
Jun 2023	0.158	0.250	**0.238**	0.300	**0.212**	**ML; Kalshi 10d**
Jul 2023	0.160	**0.180**	0.266	**0.200**	**0.160**	**Bottom-up; Consensus; Kalshi 10d**
Aug 2023	0.278	0.200	**0.275**	0.200	**0.262**	**ML; Kalshi 10d**
Sep 2023	0.323	0.200	**0.274**	**0.300**	0.176	**ML; Consensus**
Oct 2023	0.227	0.370	**0.254**	**0.300**	**0.308**	**ML; Consensus; Kalshi 10d**
Nov 2023	0.285	**0.340**	**0.344**	**0.300**	**0.342**	**Bottom-up; ML; Consensus; Kalshi 10d**
Dec 2023	0.309	**0.290**	**0.254**	**0.300**	0.242	**Bottom-up; ML; Consensus**
Jan 2024	0.392	0.230	0.199	**0.300**	0.231	**Consensus**
Feb 2024	0.358	**0.340**	**0.262**	**0.300**	**0.305**	**Bottom-up; ML; Consensus; Kalshi 10d**
Mar 2024	0.359	**0.330**	0.240	**0.300**	**0.269**	**Bottom-up; Consensus; Kalshi 10d**
Apr 2024	0.292	**0.300**	**0.268**	**0.300**	**0.322**	**Bottom-up; ML; Consensus; Kalshi 10d**
May 2024	0.163	0.280	**0.241**	0.300	0.301	**ML**
Jun 2024	0.065	**0.230**	**0.164**	**0.200**	**0.175**	**Bottom-up; ML; Consensus; Kalshi 10d**
Jul 2024	0.165	**0.210**	**0.200**	**0.200**	0.327	**Bottom-up; ML; Consensus**
Aug 2024	0.281	**0.240**	**0.232**	**0.200**	**0.209**	**Bottom-up; ML; Consensus; Kalshi 10d**
Sep 2024	0.312	**0.250**	0.237	0.200	0.208	**Bottom-up**
Oct 2024	0.280	**0.320**	**0.280**	**0.300**	**0.348**	**Bottom-up; ML; Consensus; Kalshi 10d**
Nov 2024	0.308	**0.300**	0.219	**0.300**	**0.281**	**Bottom-up; Consensus; Kalshi 10d**
Dec 2024	0.225	0.260	0.259	**0.200**	0.279	**Consensus**
Jan 2025	0.446	**0.330**	0.162	**0.300**	**0.304**	**Bottom-up; Consensus; Kalshi 10d**
Feb 2025	0.227	**0.300**	**0.305**	**0.300**	**0.301**	**Bottom-up; ML; Consensus; Kalshi 10d**
Mar 2025	0.057	**0.260**	**0.306**	**0.300**	**0.254**	**Bottom-up; ML; Consensus; Kalshi 10d**
Apr 2025	0.237	0.300	**0.224**	0.300	0.303	**ML**
May 2025	0.130	0.270	**0.168**	0.300	0.275	**ML**
Jun 2025	0.228	**0.243**	**0.225**	0.300	**0.236**	**Bottom-up; ML; Kalshi 10d**
Jul 2025	0.322	**0.310**	**0.258**	**0.300**	**0.310**	**Bottom-up; ML; Consensus; Kalshi 10d**
Aug 2025	0.346	**0.280**	**0.301**	**0.300**	**0.329**	**Bottom-up; ML; Consensus; Kalshi 10d**
Dec 2025	0.239	0.400	**0.271**	**0.300**	**0.271**	**ML; Consensus; Kalshi 10d**
Jan 2026	0.295	**0.330**	**0.311**	**0.300**	**0.303**	**Bottom-up; ML; Consensus; Kalshi 10d**
Feb 2026	0.216	**0.220**	0.328	**0.200**	0.256	**Bottom-up; Consensus**
Mar 2026	0.196	0.260	**0.225**	0.300	0.305	**ML**
Apr 2026	0.376	**0.360**	0.224	0.300	**0.365**	**Bottom-up; Kalshi 10d**
May 2026	0.210	**0.210**	0.299	0.300	0.257	**Bottom-up**
Jun 2026	-0.020	**0.230**	**0.244**	**0.200**	**0.180**	**Bottom-up; ML; Consensus; Kalshi 10d**

Values are percentage points and are displayed to three decimals. Winner determination uses one-decimal displayed values, not the extra digits shown for auditability.

Core YoY - monthly forecasts (N=39).

Reference month	Published	Bottom-up	ML	Consensus	Kalshi 10d	Winner(s)
Jan 2023	5.540	**5.488**	**5.522**	**5.500**	5.903	**Bottom-up; ML; Consensus**
Feb 2023	5.498	5.446	**5.452**	**5.500**	**5.474**	**ML; Consensus; Kalshi 10d**
Mar 2023	5.567	**5.622**	5.538	**5.600**	4.968	**Bottom-up; Consensus**
Apr 2023	5.514	5.585	5.354	**5.500**	5.142	**Consensus**
May 2023	5.338	**5.272**	5.227	**5.300**	4.603	**Bottom-up; Consensus**
Jun 2023	4.854	**4.946**	**4.938**	5.000	**4.865**	**Bottom-up; ML; Kalshi 10d**
Jul 2023	4.701	**4.721**	4.812	4.800	4.622	**Bottom-up**
Aug 2023	4.399	4.321	**4.395**	4.300	4.250	**ML**
Sep 2023	4.129	4.006	**4.078**	**4.100**	4.184	**ML; Consensus**
Oct 2023	4.027	4.171	**4.056**	**4.100**	3.679	**ML; Consensus**
Nov 2023	4.018	4.073	4.079	**4.000**	3.721	**Consensus**
Dec 2023	3.915	**3.896**	**3.858**	3.800	**3.906**	**Bottom-up; ML; Kalshi 10d**
Jan 2024	3.858	3.696	3.658	3.700	**3.752**	**Kalshi 10d**
Feb 2024	3.762	**3.744**	**3.662**	**3.700**	**3.737**	**Bottom-up; ML; Consensus; Kalshi 10d**
Mar 2024	3.816	**3.787**	3.693	3.700	3.674	**Bottom-up**
Apr 2024	3.626	**3.634**	**3.601**	**3.600**	3.653	**Bottom-up; ML; Consensus**
May 2024	3.394	**3.511**	**3.475**	**3.500**	3.554	**Bottom-up; ML; Consensus**
Jun 2024	3.265	**3.431**	**3.368**	**3.400**	3.475	**Bottom-up; ML; Consensus**
Jul 2024	3.230	3.275	3.266	**3.200**	**3.196**	**Consensus; Kalshi 10d**
Aug 2024	3.285	**3.244**	**3.235**	**3.200**	3.124	**Bottom-up; ML; Consensus**
Sep 2024	3.282	**3.220**	**3.204**	**3.200**	**3.201**	**Bottom-up; ML; Consensus; Kalshi 10d**
Oct 2024	3.299	**3.338**	**3.299**	**3.300**	3.047	**Bottom-up; ML; Consensus**
Nov 2024	3.286	**3.278**	3.194	**3.300**	**3.314**	**Bottom-up; Consensus; Kalshi 10d**
Dec 2024	3.213	**3.248**	**3.248**	3.300	3.288	**Bottom-up; ML**
Jan 2025	3.283	**3.167**	2.991	3.100	**3.151**	**Bottom-up; Kalshi 10d**
Feb 2025	3.138	**3.211**	**3.218**	**3.200**	**3.179**	**Bottom-up; ML; Consensus; Kalshi 10d**
Mar 2025	2.809	**3.012**	3.065	**3.000**	**3.050**	**Bottom-up; Consensus; Kalshi 10d**
Apr 2025	2.776	**2.839**	**2.763**	**2.800**	2.879	**Bottom-up; ML; Consensus**
May 2025	2.766	2.906	**2.805**	2.900	2.930	**ML**
Jun 2025	2.911	**2.926**	**2.907**	3.000	**2.938**	**Bottom-up; ML; Kalshi 10d**
Jul 2025	3.054	**3.042**	**2.988**	**3.000**	**3.035**	**Bottom-up; ML; Consensus; Kalshi 10d**
Aug 2025	3.112	3.046	**3.065**	**3.100**	**3.082**	**ML; Consensus; Kalshi 10d**
Dec 2025	2.646	2.807	**2.679**	**2.700**	**2.709**	**ML; Consensus; Kalshi 10d**
Jan 2026	2.512	**2.547**	**2.529**	**2.500**	2.586	**Bottom-up; ML; Consensus**
Feb 2026	2.472	**2.476**	2.587	**2.500**	**2.517**	**Bottom-up; Consensus; Kalshi 10d**
Mar 2026	2.602	2.666	**2.632**	2.700	2.673	**ML**
Apr 2026	2.743	**2.727**	2.588	**2.700**	**2.708**	**Bottom-up; Consensus; Kalshi 10d**
May 2026	2.823	**2.823**	2.915	2.900	2.897	**Bottom-up**
Jun 2026	2.566	**2.816**	**2.837**	**2.800**	**2.794**	**Bottom-up; ML; Consensus; Kalshi 10d**

Values are percentage points and are displayed to three decimals. Winner determination uses one-decimal displayed values, not the extra digits shown for auditability.

Appendix C: Monthly forecast ledger - 1h

Each table reports the published release and every forecast used in the 1h comparison. A shaded, bold forecast is a row winner: it has the smallest absolute miss after both the forecast and the release are rounded to one decimal. Ties are retained. The final column names every winner explicitly.

Headline MoM - monthly forecasts (N=34).

Reference month	Published	Bottom-up	ML	Consensus	Kalshi 1h	Winner(s)
Jun 2023	0.180	0.260	**0.239**	0.300	**0.193**	**ML; Kalshi 1h**
Jul 2023	0.167	**0.160**	**0.243**	**0.200**	**0.206**	**Bottom-up; ML; Consensus; Kalshi 1h**
Aug 2023	0.631	**0.640**	**0.555**	**0.600**	**0.639**	**Bottom-up; ML; Consensus; Kalshi 1h**
Sep 2023	0.396	0.240	**0.325**	**0.300**	**0.331**	**ML; Consensus; Kalshi 1h**
Oct 2023	0.045	**0.140**	**0.067**	**0.100**	**0.089**	**Bottom-up; ML; Consensus; Kalshi 1h**
Nov 2023	0.097	**0.110**	**0.146**	**0.100**	**0.063**	**Bottom-up; ML; Consensus; Kalshi 1h**
Dec 2023	0.303	**0.250**	0.203	0.200	0.243	**Bottom-up**
Jan 2024	0.305	**0.150**	0.115	**0.200**	**0.163**	**Bottom-up; Consensus; Kalshi 1h**
Feb 2024	0.442	**0.420**	0.347	**0.400**	0.453	**Bottom-up; Consensus**
Mar 2024	0.378	**0.350**	**0.365**	0.300	0.292	**Bottom-up; ML**
Apr 2024	0.313	0.390	**0.267**	0.400	0.374	**ML**
May 2024	0.006	**0.140**	**0.133**	**0.100**	**0.116**	**Bottom-up; ML; Consensus; Kalshi 1h**
Jun 2024	-0.056	0.090	**0.015**	0.100	**0.045**	**ML; Kalshi 1h**
Jul 2024	0.155	**0.210**	**0.185**	**0.200**	**0.177**	**Bottom-up; ML; Consensus; Kalshi 1h**
Aug 2024	0.187	**0.180**	**0.185**	**0.200**	**0.167**	**Bottom-up; ML; Consensus; Kalshi 1h**
Sep 2024	0.180	**0.090**	**0.118**	**0.100**	**0.097**	**Bottom-up; ML; Consensus; Kalshi 1h**
Oct 2024	0.244	0.250	**0.216**	**0.200**	0.374	**ML; Consensus**
Nov 2024	0.313	**0.300**	0.242	**0.300**	**0.291**	**Bottom-up; Consensus; Kalshi 1h**
Dec 2024	0.393	**0.350**	**0.364**	0.300	**0.374**	**Bottom-up; ML; Kalshi 1h**
Jan 2025	0.467	**0.350**	0.225	0.300	0.317	**Bottom-up**
Feb 2025	0.216	**0.250**	**0.279**	**0.300**	**0.305**	**Bottom-up; ML; Consensus; Kalshi 1h**
Mar 2025	-0.050	**0.130**	0.172	**0.100**	**0.071**	**Bottom-up; Consensus; Kalshi 1h**
Apr 2025	0.221	**0.310**	**0.147**	**0.300**	**0.270**	**Bottom-up; ML; Consensus; Kalshi 1h**
May 2025	0.081	0.190	**0.110**	0.200	0.176	**ML**
Jun 2025	0.287	0.230	0.218	**0.300**	0.244	**Consensus**
Jul 2025	0.197	**0.220**	**0.208**	**0.200**	0.259	**Bottom-up; ML; Consensus**
Aug 2025	0.382	**0.320**	**0.324**	**0.300**	**0.321**	**Bottom-up; ML; Consensus; Kalshi 1h**
Dec 2025	0.307	0.370	**0.251**	**0.300**	**0.320**	**ML; Consensus; Kalshi 1h**
Jan 2026	0.171	**0.220**	**0.193**	0.300	**0.241**	**Bottom-up; ML; Kalshi 1h**
Feb 2026	0.267	**0.270**	0.354	**0.300**	**0.273**	**Bottom-up; Consensus; Kalshi 1h**
Mar 2026	0.865	**0.910**	0.976	**0.900**	**0.900**	**Bottom-up; Consensus; Kalshi 1h**
Apr 2026	0.640	**0.610**	0.522	**0.600**	**0.594**	**Bottom-up; Consensus; Kalshi 1h**
May 2026	0.470	**0.520**	0.626	**0.500**	**0.496**	**Bottom-up; Consensus; Kalshi 1h**
Jun 2026	-0.420	**-0.180**	**-0.164**	-0.100	-0.124	**Bottom-up; ML**

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Headline YoY - monthly forecasts (N=35).

Reference month	Published	Bottom-up	ML	Consensus	Kalshi 1h	Winner(s)
Jun 2023	3.071	3.150	**3.131**	**3.100**	**3.080**	**ML; Consensus; Kalshi 1h**
Jul 2023	3.288	**3.281**	3.367	**3.300**	3.243	**Bottom-up; Consensus**
Aug 2023	3.723	**3.731**	3.644	3.600	**3.664**	**Bottom-up; Kalshi 1h**
Sep 2023	3.687	3.531	**3.614**	**3.600**	**3.627**	**ML; Consensus; Kalshi 1h**
Oct 2023	3.251	**3.346**	**3.274**	**3.300**	3.425	**Bottom-up; ML; Consensus**
Nov 2023	3.133	**3.146**	3.184	**3.100**	**3.104**	**Bottom-up; Consensus; Kalshi 1h**
Dec 2023	3.316	**3.263**	3.213	3.200	**3.329**	**Bottom-up; Kalshi 1h**
Jan 2024	3.088	2.933	2.892	2.900	**2.989**	**Kalshi 1h**
Feb 2024	3.157	3.135	3.060	3.100	**3.156**	**Kalshi 1h**
Mar 2024	3.487	**3.459**	**3.474**	3.400	3.375	**Bottom-up; ML**
Apr 2024	3.361	**3.438**	3.314	**3.400**	3.458	**Bottom-up; Consensus**
May 2024	3.244	**3.379**	**3.376**	**3.400**	**3.371**	**Bottom-up; ML; Consensus; Kalshi 1h**
Jun 2024	2.970	3.116	**3.043**	3.100	3.102	**ML**
Jul 2024	2.941	**2.997**	**2.973**	**3.000**	3.322	**Bottom-up; ML; Consensus**
Aug 2024	2.607	**2.600**	**2.605**	**2.600**	2.526	**Bottom-up; ML; Consensus**
Sep 2024	2.426	2.337	**2.363**	2.300	**2.378**	**ML; Kalshi 1h**
Oct 2024	2.579	**2.585**	**2.551**	**2.600**	**2.638**	**Bottom-up; ML; Consensus; Kalshi 1h**
Nov 2024	2.719	**2.707**	2.647	**2.700**	**2.737**	**Bottom-up; Consensus; Kalshi 1h**
Dec 2024	2.871	2.828	2.841	**2.900**	**2.896**	**Consensus; Kalshi 1h**
Jan 2025	2.991	**2.874**	2.743	**2.900**	**2.928**	**Bottom-up; Consensus; Kalshi 1h**
Feb 2025	2.802	**2.836**	2.866	2.900	2.905	**Bottom-up**
Mar 2025	2.382	2.562	2.609	2.600	**2.521**	**Kalshi 1h**
Apr 2025	2.325	2.414	**2.250**	2.400	2.389	**ML**
May 2025	2.377	2.486	**2.407**	2.500	**2.431**	**ML; Kalshi 1h**
Jun 2025	2.680	2.623	2.610	**2.700**	2.620	**Consensus**
Jul 2025	2.743	**2.766**	**2.754**	**2.800**	**2.795**	**Bottom-up; ML; Consensus; Kalshi 1h**
Aug 2025	2.939	**2.876**	**2.878**	**2.900**	**2.880**	**Bottom-up; ML; Consensus; Kalshi 1h**
Nov 2025	2.696	**2.802**	**2.778**	3.100	3.001	**Bottom-up; ML**
Dec 2025	2.653	**2.716**	2.596	**2.700**	**2.729**	**Bottom-up; Consensus; Kalshi 1h**
Jan 2026	2.391	**2.440**	**2.414**	2.500	2.493	**Bottom-up; ML**
Feb 2026	2.434	**2.437**	2.523	**2.400**	2.462	**Bottom-up; Consensus**
Mar 2026	3.286	**3.331**	3.399	**3.300**	**3.337**	**Bottom-up; Consensus; Kalshi 1h**
Apr 2026	3.779	**3.749**	**3.657**	**3.700**	**3.708**	**Bottom-up; ML; Consensus; Kalshi 1h**
May 2026	4.167	**4.217**	4.329	**4.200**	**4.235**	**Bottom-up; Consensus; Kalshi 1h**
Jun 2026	3.464	**3.704**	**3.729**	3.800	3.808	**Bottom-up; ML**

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Core MoM - monthly forecasts (N=39).

Reference month	Published	Bottom-up	ML	Consensus	Kalshi 1h	Winner(s)
Jan 2023	0.412	**0.360**	**0.396**	**0.400**	**0.385**	Bottom-up; ML; Consensus; Kalshi 1h
Feb 2023	0.452	**0.400**	**0.408**	**0.400**	**0.398**	Bottom-up; ML; Consensus; Kalshi 1h
Mar 2023	0.385	**0.440**	**0.358**	**0.400**	**0.435**	Bottom-up; ML; Consensus; Kalshi 1h
Apr 2023	0.409	0.480	0.256	**0.400**	0.451	Consensus
May 2023	0.436	**0.370**	0.330	**0.400**	0.333	Bottom-up; Consensus
Jun 2023	0.158	0.250	**0.238**	0.300	**0.204**	ML; Kalshi 1h
Jul 2023	0.160	**0.180**	0.266	**0.200**	**0.157**	Bottom-up; Consensus; Kalshi 1h
Aug 2023	0.278	0.200	**0.275**	0.200	**0.258**	ML; Kalshi 1h
Sep 2023	0.323	0.200	**0.274**	**0.300**	**0.304**	ML; Consensus; Kalshi 1h
Oct 2023	0.227	0.370	**0.254**	**0.300**	**0.346**	ML; Consensus; Kalshi 1h
Nov 2023	0.285	**0.340**	**0.344**	**0.300**	0.247	Bottom-up; ML; Consensus
Dec 2023	0.309	**0.290**	**0.254**	**0.300**	**0.257**	Bottom-up; ML; Consensus; Kalshi 1h
Jan 2024	0.392	0.230	0.199	**0.300**	**0.279**	Consensus; Kalshi 1h
Feb 2024	0.358	0.340	0.262	0.300	**0.380**	Kalshi 1h
Mar 2024	0.359	**0.330**	0.240	**0.300**	**0.281**	Bottom-up; Consensus; Kalshi 1h
Apr 2024	0.292	**0.300**	**0.268**	**0.300**	**0.309**	Bottom-up; ML; Consensus; Kalshi 1h
May 2024	0.163	0.280	**0.241**	0.300	0.277	ML
Jun 2024	0.065	**0.230**	**0.164**	**0.200**	**0.189**	Bottom-up; ML; Consensus; Kalshi 1h
Jul 2024	0.165	**0.210**	**0.200**	**0.200**	0.333	Bottom-up; ML; Consensus
Aug 2024	0.281	**0.240**	**0.232**	**0.200**	**0.215**	Bottom-up; ML; Consensus; Kalshi 1h
Sep 2024	0.312	**0.250**	0.237	0.200	**0.257**	Bottom-up; Kalshi 1h
Oct 2024	0.280	**0.320**	**0.280**	**0.300**	0.351	Bottom-up; ML; Consensus
Nov 2024	0.308	**0.300**	0.219	**0.300**	**0.288**	Bottom-up; Consensus; Kalshi 1h
Dec 2024	0.225	0.260	0.259	**0.200**	0.276	Consensus
Jan 2025	0.446	**0.330**	0.162	**0.300**	**0.307**	Bottom-up; Consensus; Kalshi 1h
Feb 2025	0.227	**0.300**	**0.305**	**0.300**	**0.315**	Bottom-up; ML; Consensus; Kalshi 1h
Mar 2025	0.057	0.260	0.306	0.300	**0.239**	Kalshi 1h
Apr 2025	0.237	0.300	**0.224**	0.300	0.280	ML
May 2025	0.130	0.270	**0.168**	0.300	0.267	ML
Jun 2025	0.228	**0.243**	**0.225**	0.300	0.295	Bottom-up; ML
Jul 2025	0.322	**0.310**	**0.258**	**0.300**	**0.322**	Bottom-up; ML; Consensus; Kalshi 1h
Aug 2025	0.346	**0.280**	**0.301**	**0.300**	**0.308**	Bottom-up; ML; Consensus; Kalshi 1h
Dec 2025	0.239	0.400	**0.271**	**0.300**	**0.326**	ML; Consensus; Kalshi 1h
Jan 2026	0.295	**0.330**	**0.311**	**0.300**	**0.328**	Bottom-up; ML; Consensus; Kalshi 1h
Feb 2026	0.216	**0.220**	0.328	**0.200**	**0.241**	Bottom-up; Consensus; Kalshi 1h
Mar 2026	0.196	0.260	**0.225**	0.300	0.262	ML
Apr 2026	0.376	**0.360**	0.224	0.300	**0.376**	Bottom-up; Kalshi 1h
May 2026	0.210	**0.210**	0.299	0.300	**0.240**	Bottom-up; Kalshi 1h
Jun 2026	-0.020	**0.230**	**0.244**	**0.200**	**0.231**	Bottom-up; ML; Consensus; Kalshi 1h

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Core YoY - monthly forecasts (N=39).

Reference month	Published	Bottom-up	ML	Consensus	Kalshi 1h	Winner(s)
Jan 2023	5.540	**5.488**	**5.522**	**5.500**	5.650	**Bottom-up; ML; Consensus**
Feb 2023	5.498	5.446	**5.452**	**5.500**	**5.466**	**ML; Consensus; Kalshi 1h**
Mar 2023	5.567	**5.622**	5.538	**5.600**	5.170	**Bottom-up; Consensus**
Apr 2023	5.514	5.585	5.354	**5.500**	5.174	**Consensus**
May 2023	5.338	**5.272**	5.227	**5.300**	4.754	**Bottom-up; Consensus**
Jun 2023	4.854	**4.946**	**4.938**	5.000	4.813	**Bottom-up; ML**
Jul 2023	4.701	**4.721**	4.812	4.800	4.629	**Bottom-up**
Aug 2023	4.399	4.321	**4.395**	4.300	4.268	**ML**
Sep 2023	4.129	4.006	**4.078**	**4.100**	4.392	**ML; Consensus**
Oct 2023	4.027	4.171	**4.056**	**4.100**	3.681	**ML; Consensus**
Nov 2023	4.018	4.073	4.079	**4.000**	**4.002**	**Consensus; Kalshi 1h**
Dec 2023	3.915	**3.896**	**3.858**	3.800	4.173	**Bottom-up; ML**
Jan 2024	3.858	3.696	3.658	3.700	**3.796**	**Kalshi 1h**
Feb 2024	3.762	3.744	3.662	3.700	**3.757**	**Kalshi 1h**
Mar 2024	3.816	**3.787**	3.693	3.700	3.700	**Bottom-up**
Apr 2024	3.626	**3.634**	**3.601**	**3.600**	3.656	**Bottom-up; ML; Consensus**
May 2024	3.394	**3.511**	**3.475**	**3.500**	**3.522**	**Bottom-up; ML; Consensus; Kalshi 1h**
Jun 2024	3.265	**3.431**	**3.368**	**3.400**	**3.393**	**Bottom-up; ML; Consensus; Kalshi 1h**
Jul 2024	3.230	3.275	3.266	**3.200**	**3.208**	**Consensus; Kalshi 1h**
Aug 2024	3.285	**3.244**	**3.235**	**3.200**	3.100	**Bottom-up; ML; Consensus**
Sep 2024	3.282	3.220	3.204	3.200	**3.256**	**Kalshi 1h**
Oct 2024	3.299	**3.338**	**3.299**	**3.300**	3.047	**Bottom-up; ML; Consensus**
Nov 2024	3.286	**3.278**	3.194	**3.300**	**3.312**	**Bottom-up; Consensus; Kalshi 1h**
Dec 2024	3.213	**3.248**	**3.248**	3.300	3.268	**Bottom-up; ML**
Jan 2025	3.283	**3.167**	2.991	3.100	3.111	**Bottom-up**
Feb 2025	3.138	**3.211**	**3.218**	**3.200**	**3.227**	**Bottom-up; ML; Consensus; Kalshi 1h**
Mar 2025	2.809	**3.012**	3.065	**3.000**	3.118	**Bottom-up; Consensus**
Apr 2025	2.776	**2.839**	**2.763**	**2.800**	**2.843**	**Bottom-up; ML; Consensus; Kalshi 1h**
May 2025	2.766	2.906	**2.805**	2.900	3.000	**ML**
Jun 2025	2.911	**2.926**	**2.907**	3.000	**2.908**	**Bottom-up; ML; Kalshi 1h**
Jul 2025	3.054	3.042	2.988	3.000	**3.050**	**Kalshi 1h**
Aug 2025	3.112	3.046	**3.065**	**3.100**	**3.074**	**ML; Consensus; Kalshi 1h**
Dec 2025	2.646	2.807	**2.679**	**2.700**	2.766	**ML; Consensus**
Jan 2026	2.512	**2.547**	**2.529**	**2.500**	2.587	**Bottom-up; ML; Consensus**
Feb 2026	2.472	**2.476**	2.587	**2.500**	**2.490**	**Bottom-up; Consensus; Kalshi 1h**
Mar 2026	2.602	2.666	**2.632**	2.700	2.656	**ML**
Apr 2026	2.743	**2.727**	2.588	**2.700**	**2.723**	**Bottom-up; Consensus; Kalshi 1h**
May 2026	2.823	**2.823**	2.915	2.900	**2.832**	**Bottom-up; Kalshi 1h**
Jun 2026	2.566	**2.816**	**2.837**	**2.800**	**2.768**	**Bottom-up; ML; Consensus; Kalshi 1h**

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